

Interference

1. Introduction

In the 17th century, the properties of light were explained by Sir Isaac Newton and Christian Huygens. Sir Isaac Newton was explained the properties of light by introducing Corpuscular theory in 1675. It explains reflection, refraction, and dispersion properties of light. It fails to explain interference, diffraction, polarization, photo electric effect, and double refraction.

In 1679, Christian Huygens proposed the wave theory of light. According to Huygens wave theory, each point on the wave front is to be considered as a source of secondary wavelets. It explains reflection, refraction, dispersion, double refraction, diffraction, interference, and polarization properties of light. It fails to explain, photo electric effect, black body radiation etc,

2. Interference of light

The best evidence for the wave nature of light is interference phenomenon. This was experimentally demonstrated by Thomas Young in 180, through double slit experiment. Due to interference, we will observe many observations in our day today life, such as multiple colours on soap bubbles as well as on oil film when viewed under sun light. Interference concept is explained on the basis of superposition of wave's concept.

When two light waves superimpose, then the resultant amplitude or intensity in the region of superposition is different than the amplitude of individual waves.

Definition:-

The modification in the distribution of intensity in the region of superposition is known as interference.

In case of interference pattern we observe two cases

- Constructive interference
- Destructive interference

Constructive interference

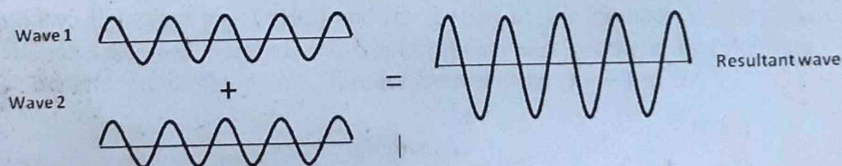
➤ The waves are reaching at a point are in phase constructive interference occurs

➤ In constructive interference, the resultant amplitude is always equal to the sum of the amplitudes of two individual waves.

➤ Condition

The path difference between the two waves is equal to the integral multiple of wave length (λ) the constructive interference occurs.

path difference = $n\lambda$ Where $n = 0, 1, 2, 3, 4 \dots$



Destructive interference

➤ The waves are reaching at a point are in out of phase destructive interference occurs

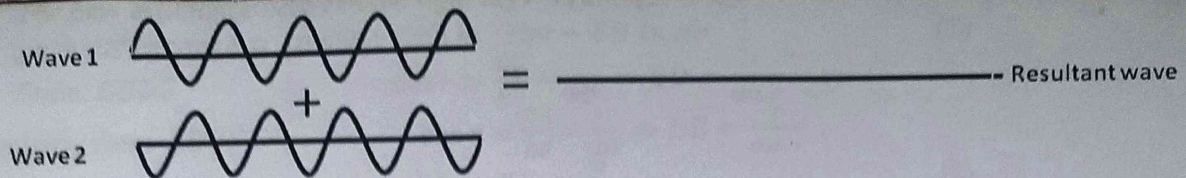
➤ In Destructive interference, the resultant amplitude is always equal to the difference of the amplitudes of two individual waves.

➤ Condition

The path difference between the two waves is equal to the odd integral multiple of $\lambda/2$ destructive interference occurs

path difference = $\frac{(2n-1)\lambda}{2}$ Where $n = 1, 2, 3, 4 \dots$ or

path difference = $\frac{(2n+1)\lambda}{2}$ Where $n = 0, 1, 2, 3, 4 \dots$



3. Types of interference:-

For the formation of interference pattern, two coherent light sources are required. To get two coherent sources from a single light source, two techniques are used. They are

1. Division of wave front
2. Division of amplitude

Division of wave front

The wave front from a single light source is divided into two parts using the phenomenon of reflection, refraction, or diffraction. Young's double slit experiment is belongs to this class of interference.

Division of amplitude

The amplitude of a single light beam is divided into two parts by parallel reflection or refraction. Newton's ring experiment, Michelson's interferometer is belongs to this class of interference.

4. Conditions for interference

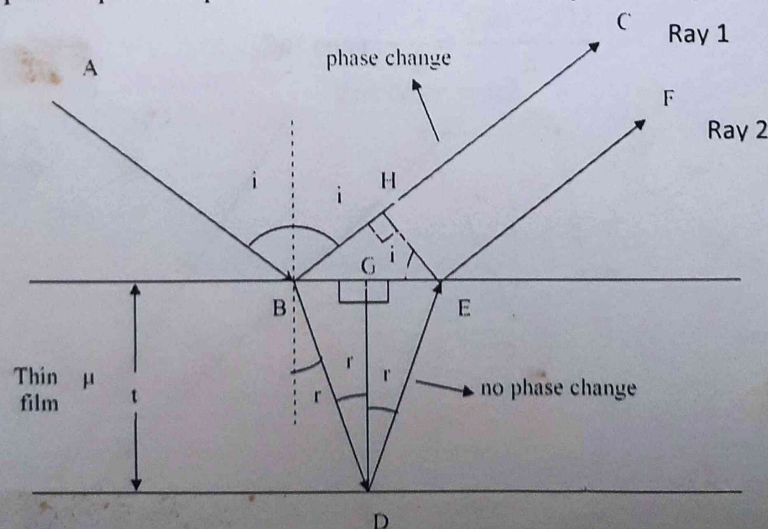
- 1) Two light sources of emitting light waves should be coherent.
- 2) Two sources must emit continuous light waves of same wavelengths or frequency.
- 3) The separation between the two sources should be small.
- 4) The distance between the two sources and the screen should be large.
- 5) To view interference fringes, the background should be dark.
- 6) The amplitude of light waves should be equal or nearly equal.
- 7) The sources should be narrow.
- 8) The sources should be monochromatic.

5. Interference in thin films by reflection

Principle:-

The formation of colours in thin films can explained as due to the phenomenon of interference. In this example, the formation of interference pattern is by the division of amplitude.

Consider a thin film of uniform thickness ' t ' and refractive index ' μ '. Let a monochromatic light ray AB is incident is on the upper surface of the film at point 'A' with an angle ' i '. The incidence light ray AB is divided into two light rays ray 1 (BC) and ray 2 (EF) by the division of amplitude principle. These two light rays BC and EF are parallel and superimpose and produce interference. The intensity of interference fringe depends up on the path difference between the ray 1 and ray 2.



The path difference between the light rays (1) and (2) is
path difference = $\mu(BD + DE)$ in film – BH in air (1)

From $\triangle BDG$ $\cos r = \frac{DG}{BD} = \frac{t}{BD} \Rightarrow BD = \frac{t}{\cos r}$

Similarly from $\triangle DEG$ $\cos r = \frac{DG}{DE} = \frac{t}{DE} \Rightarrow DE = \frac{t}{\cos r}$
 $\therefore BD = DE = \frac{t}{\cos r}$ (2)

From $\triangle BEH$ $\sin i = \frac{BH}{BE} = \frac{BH}{BG+GE}$

$\therefore BH = (BG + GE) \cdot \sin i$

From $\triangle BDG$ and $\triangle DEG$ $BG = GE = t \tan r$

$BH = (2t \tan r) \cdot \sin i$

From Snell's law at point B

$$\sin i = \mu \sin r$$

$$\therefore BH = 2\mu t \tan r \cdot \sin r \quad (3)$$

Substituting the equations (2) and (3) in equation (1), we get

$$\begin{aligned} \text{Path difference} &= \frac{2\mu t}{\cos r} - 2\mu t \tan r \cdot \sin r \\ &= \frac{2\mu t}{\cos r} - 2\mu t \cdot \frac{\sin^2 r}{\cos r} \\ &= \frac{2\mu t}{\cos r} (1 - \sin^2 r) \\ &= \frac{2\mu t}{\cos r} \cos^2 r \\ &= 2\mu t \cos r \end{aligned}$$

At point B the light ray (1) is reflected at the surface of thin film (denser medium). So the light ray (1) undergoes a phase change π or an additional path difference $\lambda/2$.

$$\text{Total path difference} = 2\mu t \cos r - \frac{\lambda}{2}$$

Constructive interference (or Bright fringe)

General condition; *path difference* = $n\lambda$

$$2\mu t \cos r - \frac{\lambda}{2} = n\lambda$$

$$2\mu t \cos r = n\lambda + \frac{\lambda}{2}$$

$$2\mu t \cos r = \frac{(2n+1)\lambda}{2}$$

Destructive interference (or Dark fringe)

General condition: *path difference* = $(2n+1)\frac{\lambda}{2}$

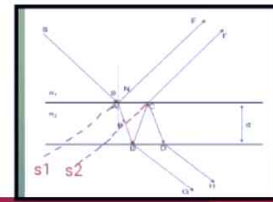
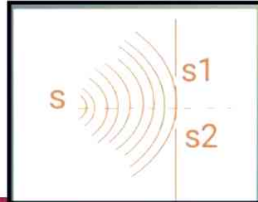
$$2\mu t \cos r - \frac{\lambda}{2} = \frac{(2n-1)\lambda}{2}$$

$$2\mu t \cos r = n\lambda$$

Formation of Colours in thin films :

- We often see bright bands of colours on the surface of a soap bubble or on a thin layer of oil floating on water.
- Normally the colours are seen in the reflected system.
- The optical path difference in the reflected system is given by $(2\mu t \cos r \pm \frac{\lambda}{2})$
- Therefore, if a film is illuminated by white light, different colours (with different wavelength) will have different optical path in the film in the given time.
- Some colours interfere constructively and due to this formation of colours take place.
- The optical path difference also depends on thickness (t) and angle of refraction (r).
- Therefore, when 't' and/or 'r' changes, optical path difference changes. This also leads to the formation of colours.

Aspect	Division of Wavefront	Division of Amplitude
Principle	Splits wavefront into parts	Splits beam by reflection/transmission
Coherent Sources	Created geometrically	Naturally coherent (from same source)
Applications	Basic optical experiments	Precision instruments & sensors
Examples	Young's slits, Biprism, Lloyd's Mirror	Newton's Rings, Michelson Interferometer



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Thin and Thick Films

Feature	Thin Film	Thick Film
Thickness Range	Nanometers to a few microns	Tens to hundreds of microns or more
Interference Effects	Prominent, visible (colors, patterns)	Negligible or minimal
Common Use	Optical, electronic, coatings	Mechanical, insulation, circuits
Examples	Soap bubbles, AR coatings, oil films	Resistors, PCBs, varnishes

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Parallel thin films and Wedge shaped films

Thin parallel film:

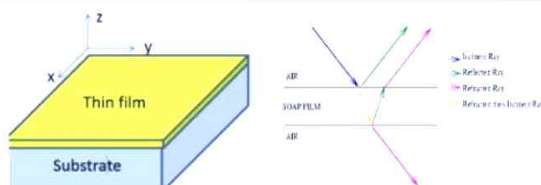
A thin film with a uniform thickness across its surface.

Applications:

Used in various applications, including anti-reflection coatings, High reflection coatings, optical filters, and mirrors.

Thickness Measurement:

The thickness can be determined by analyzing the interference patterns produced by the film.



Wedge-Shaped Thin Films:

A thin film with a thickness that varies linearly across its surface, forming a wedge shape.

Applications:

Used in applications like wedge prisms, optical components, and in the study of thin film properties.

Thickness Measurement:

The angle of the wedge and the fringe spacing can be used to determine the film's thickness at different points.

